



Adaptive Arena Sprint 3



Meet The Team



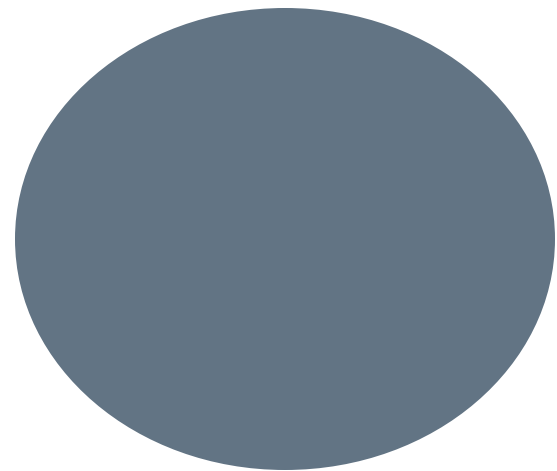
Inna Ivashchenko
Project Manager / Developer



Johnhenry Brayman
Lead Game Developer



Shariah Allen
Game Developer / AI Engineer



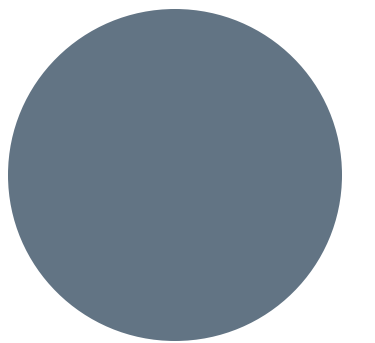
Anthony Piombino
Project Manager / QA Engineer



Gabrielle Schwan
UI/UX Developer



Henry Wong
AI/ML Advisor / Consultant





Recap sprint 2



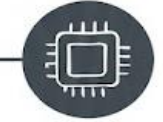
Project Description



Team Working Agreement



User Personas



MVP / Tech / Algorithms



Diagrams



Product Backlog / User Stories



Team Velocity / Burndown



Demo

Agenda

Sprint 2 Recap

Three AI difficulty levels introduced

- Easy difficulty with slower reaction
- Medium difficulty with average behavior
- Hard difficulty with machine learning to implement reward and penalty system to guide AI learning

Bug testing and resolution

Project Description

2D-fighting game focusing primarily on AI opponents and improving your general fighting game skills.

AI will scale appropriately based on selected skill level

- Easy : Mostly random or suboptimal moves
- Medium : Basic strategy with a little randomness
- Hard : Uses machine learning to adapt and respond

Provide user-friendly interface for user entry

Tally up and output results of game



Team Working Agreement

Communication And Responsibilities:

- Communication through WhatsApp, Slack, Zoom.
- All team members must respect each other and have the right to contribute their ideas to the project.
- All ideas, vision and changes discussed together, and final decisions are up to most of the team, except urgent situations.
- Each person works within the boundaries established before the start of work.
- Everyone is responsible for their own part.
- All conflicts should first be resolved between the people involved. If that is not possible the issue should be brought to the Team Lead.
- Each member must inform the team about any changes or improvements they make.

Meetings:

- Meeting happens through Zoom, Slack
- The team must have Scrum meetings two times in a week and check WhatsApp, Slack daily to avoid missing any project information.
- Schedule of a Scrum meeting : Tuesday at 8pm and Friday at 9.15pm
- Each team member should participate in at least one meeting per week unless they have an important reason.

User Personas: The Competitive Gamer

Name: Anna Grace

Age: 28

Gender: Female

Occupation: Game Developer & Twitch Streamer

Anna is a season game developer with a solid fan base, that enjoys playing games in her free time, specifically fighting, first-person shooters (FPS), and racing games. She's often searching for new and exciting games that have varying difficult levels and unique features that keeps players engaged.

Challenges:

- Tired of playing the same popular titles, with reoccurring characters and stories
- Wants to find new games developed by small teams to stream for her community

Goals:

- She enjoys playing games and learning different mechanics
- Wants to explore new games with AI elements
- Seeks a game that can be replayed several times

Quote:

“ I just want to entertain my community and learn new skills; I love a good challenge!”



User Personas: The Button Smasher

Name: Jordan Carter

Age: 19

Gender: Male

Occupation: Undergraduate Student and Student Athlete

Jordan is a busy student athlete studying animation, that loves to play games on his free time on the weekends or while hanging out with friends. He loves playing games with lots of action, unique characters, and prefers to not memorize combos but just plays for fun, not to win.

Challenges:

- Complex inputs and combinations makes gameplay frustrating
- Long or vague tutorials

Goals:

- Have fun solo or with friends
- Win occasionally, or access to varying difficulties

Quote:

“ I skip tutorials, I just want to have mindless fun!”



User Personas: The Lore Explorer

Name: Dylan Taylor

Age: 12

Gender: Male

Occupation: Middle School Student

Dylan enjoys playing games after school with friends, family, or by himself, and specifically enjoys games that are action-packed. He seeks out games that provide a good challenge, has lots of enemies, and visual stimulation. He aspires to be a game developer and enjoys learning how to make 2D games during his video game club meetings.

Challenges:

- Hard to find action games that are age appropriate
- Wants games with unique enemies
- Prefers 2D games

Goals:

- Wants to explore new action games
- Age-appropriate game with visual stimulation

Quote:

“ Games are fun when they have a lot of action!”



Inputs

Horizontal Movement:

- Move Forward: Player advances toward the opponent in the right direction using the "D" or "Right" arrow keys
- Move Backward: Player retreats from opponent towards the left direction using the "A" or "Left" arrow keys

Vertical Movement:

- Player can do a single or double jump mid-air to evade their opponent's attacks
- Uses the spacebar to do single or double jumps
 - Double tap the spacebar to do a double jump

Blocking:

- Defensive action intended to reduce or prevent incoming damage from attacks
- Use the Right Mouse Button (RMB) to block attacks



Inputs

Attacks: Kick

- A lower body attack that deals 10 damage when the player's hitbox collides with the opponent
- Press "K" to kick

Defensive: Duck

- A defensive crouch that prevents movement while active
- Cannot be performed while jumping or attacking
- Hold C to duck, release to stand

Attacks: Punch

- Basic melee attack that deals damage when the player's hitbox collides with the opponent
- Use the Left Mouse Button (LMB) to throw a punch

Attack Actions

Punching

- Triggered once an attack connects
- Characters deals 10 damage to opponent
- Opponent briefly enters a hit reaction state once hit

Kicking

- Triggered once an attack connects
- Character deals 10 damage to opponent
- Opponent briefly enters a hit reaction state once hit

Defensive Actions

Blocking

- Triggered when player holds block input
- Negates incoming damage from attacks
- Character enters blocking state

Ducking

- Triggered when player holds duck input
- Character enters ducking state
- Prevents movement while active
- Cannot be performed while jumping or attacking

Reactive Actions

Being Hit

- Triggered once an attack connects.
- Character takes damage
- May briefly enter a hit reaction state once hit
 - Each attack deals varying levels of damage
 - Ex. Punch & Kick deals 10 damage

Dying

- Occurs once a fighter's health reaches 0
- Ends Match
- Game transitions to End-Game State

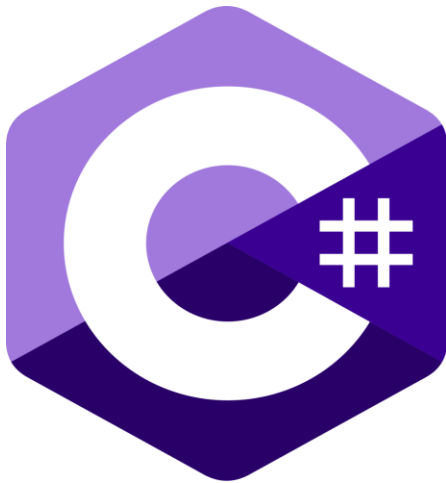
Technologies



Unity 6.3 LTS



Git LFS Extension



C# Programming



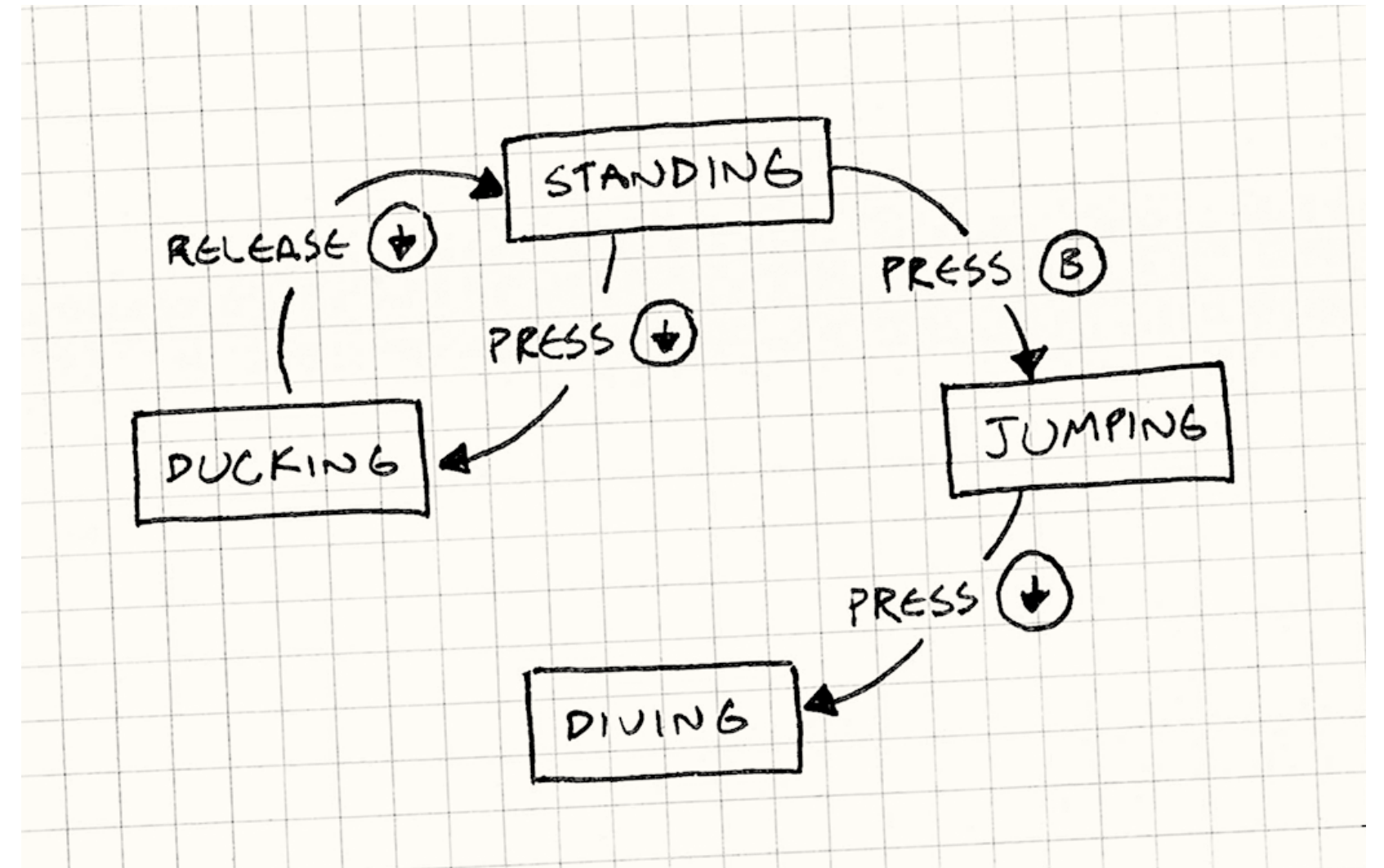
Python (for Machine Learning)



Algorithms

Easy Difficulty : Randomized Finite State Machine

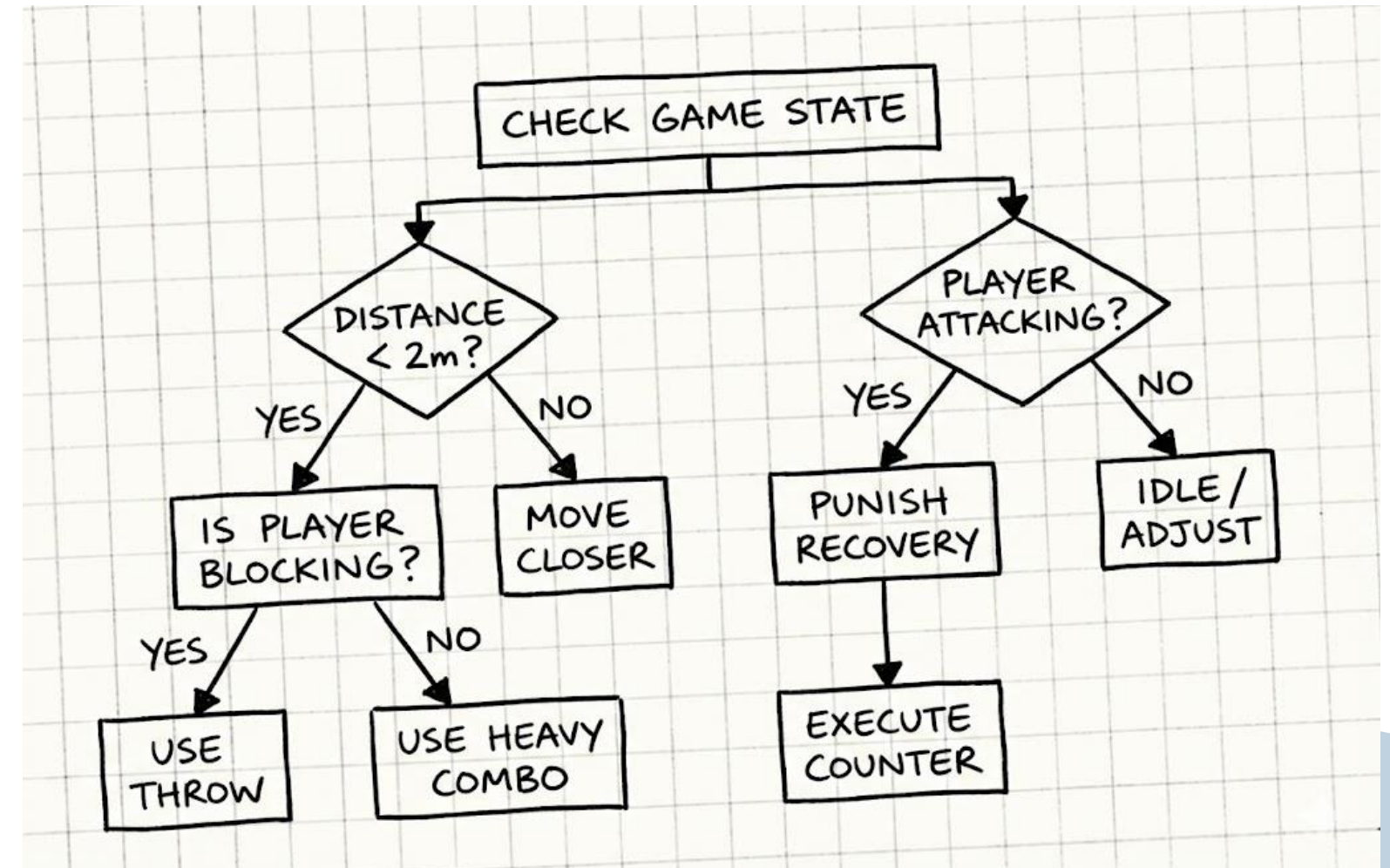
- The enemy AI is presented with multiple pre-defined states (Idle, Attack, Advance, Block, Duck).
- Instead of using strategy, it picks which state to move to next randomly.
- Attacks and movements still make sense but are suboptimal.



Algorithms

Medium Difficulty : Decision Trees

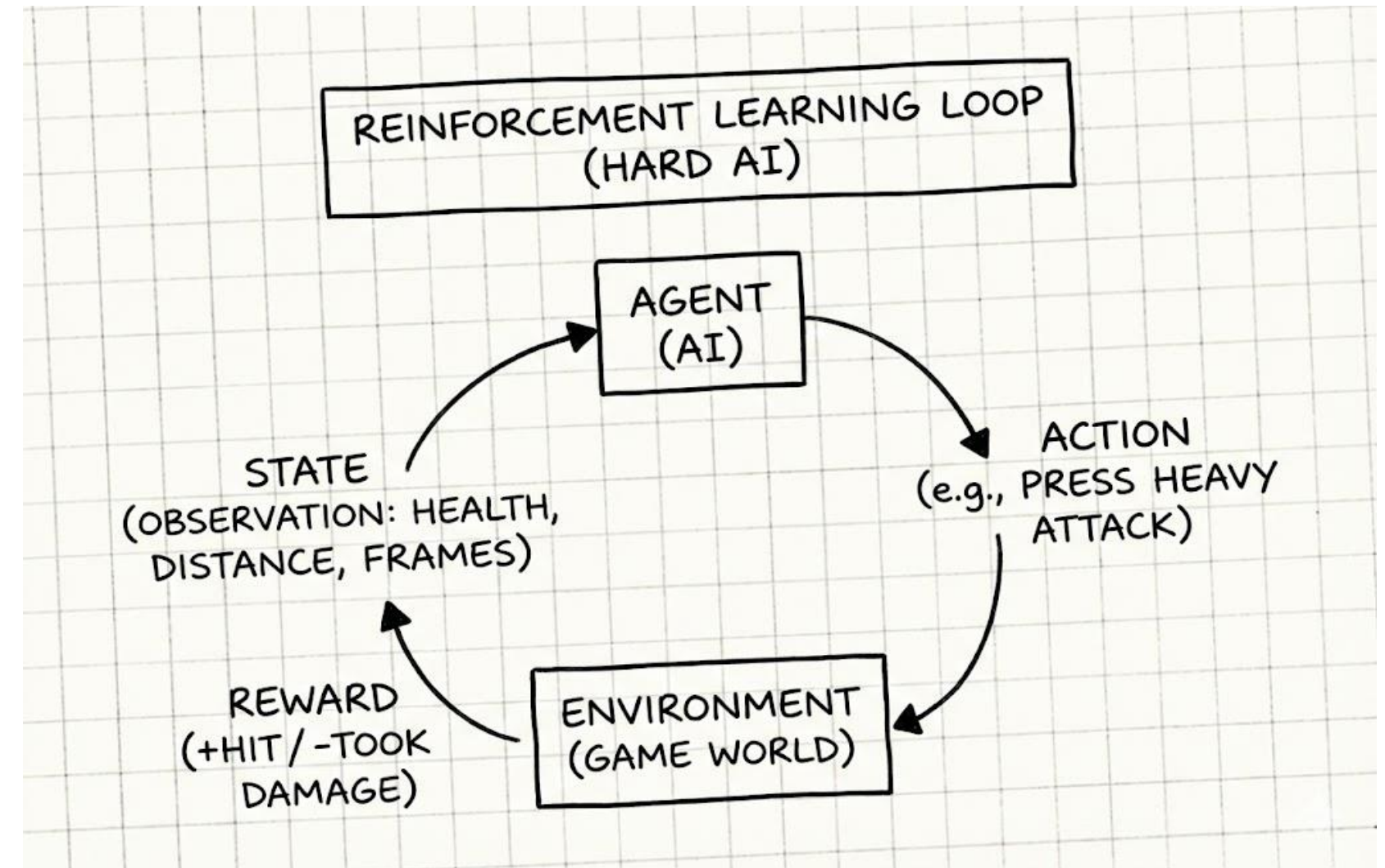
- The enemy AI in this difficulty will start to react differently depending on certain criteria.
- Before making an action, it will check a specific condition and act accordingly.
- Can still make random decisions but in a more educated way.
 - Ex. Check to see if the player is attacking. If yes, randomly choose between block or counter.



Algorithms

Hard Difficulty : Reinforcement Learning

- The enemy AI will fight through many simulations either against itself, or one of the before mentioned difficulties.
- It will receive a reward (+1) for landing a hit on the player, and a punishment (-1) for being struck by the player.
- The AI will learn which actions lead to rewards and which actions lead to punishments.



Minimum Viable Product (MVP)

The Environment:

- A simple flat ground with invisible walls on each side to prevent falling off.
- Simple environment design keeps focus on combat mechanics and player interaction
- A basic camera with a script that keeps both players in-frame while zooming in as close as possible.

The AI Difficulties:

- For the rules-based difficulties, the State Machine and Decision Tree logic must work, even if they are not complex.
- For the hard difficulty, the Unity Agent must be able to read observations and output actions from the Neural Net.

The Combat System:

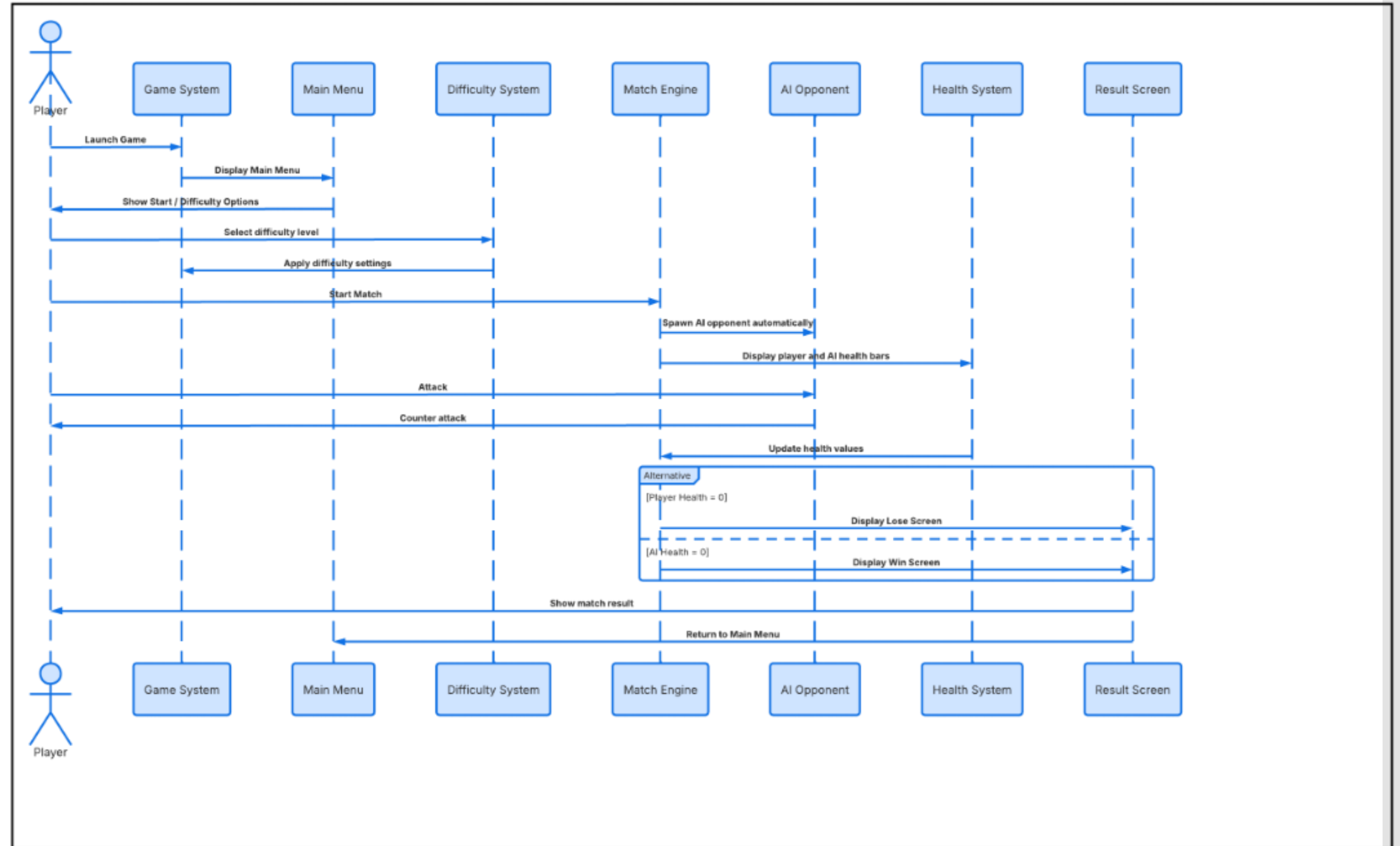
- The bare minimum controls will be Locomotion, an Attack action, and a Block action.
- Hit detection uses colliders to determine when attacks connect
- Possibility to include combos, ducking, countering, and throwing later.

The Game Flow:

- The player must be able to enter the game from a main menu and select an AI difficulty
- Loads varying end-game scenes depending on match outcome
- In-game must show health bars depleting, and a return to the main menu when one character reaches zero.

Sequence Diagram

Fighting Game Gameplay Sequence



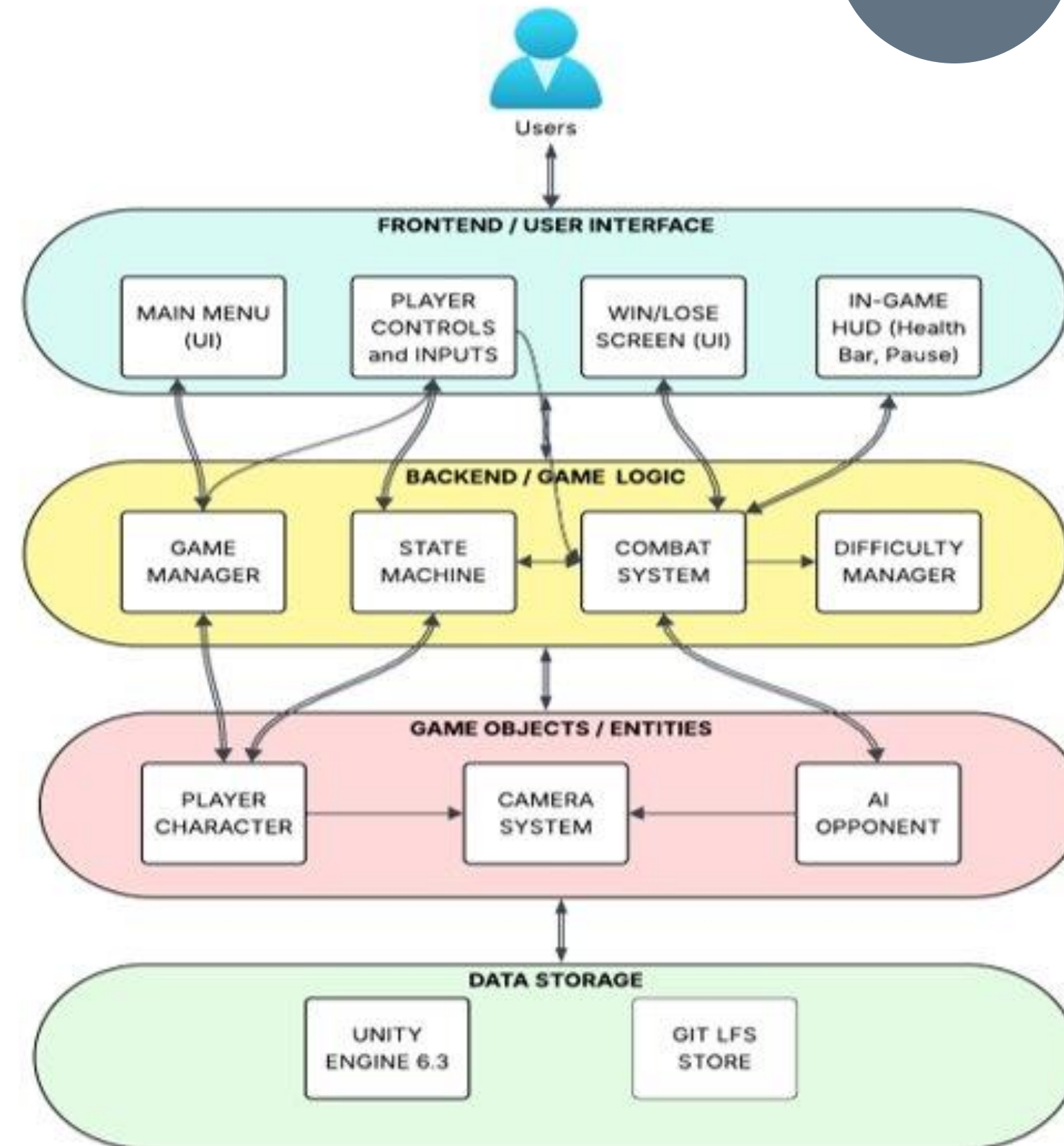
Conceptual Architecture Diagram

Frontend: Accepts player input and displays health status (HUD), Win/Lose Screen.

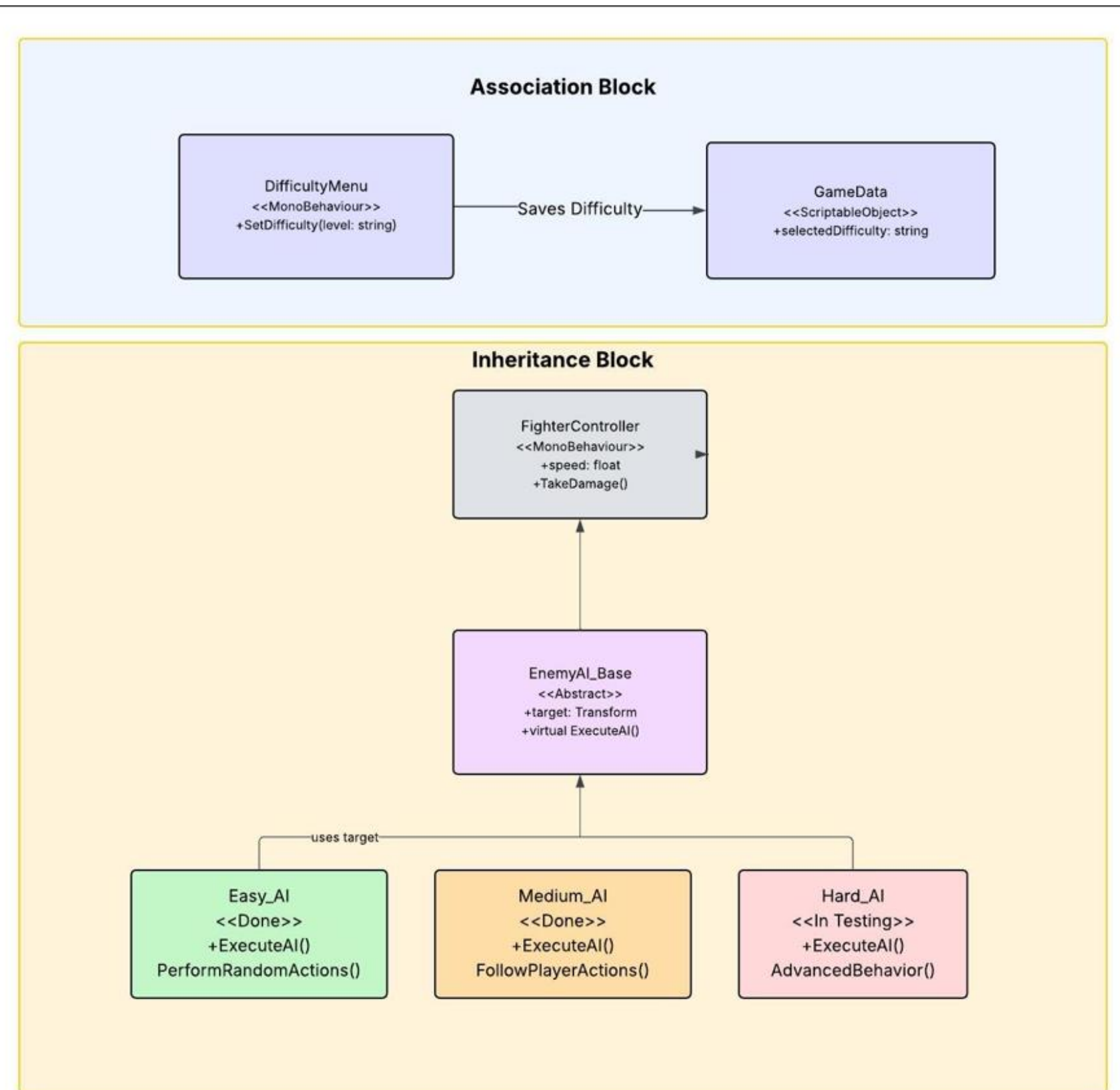
Backend: The "brain" of the game, managing combat logic, character states, and difficulty.

Entities: Game characters and a "smart" camera, operating according to the backend's rules.

Data Storage: A stable foundation built on Unity 6.3 and Git LFS storage.



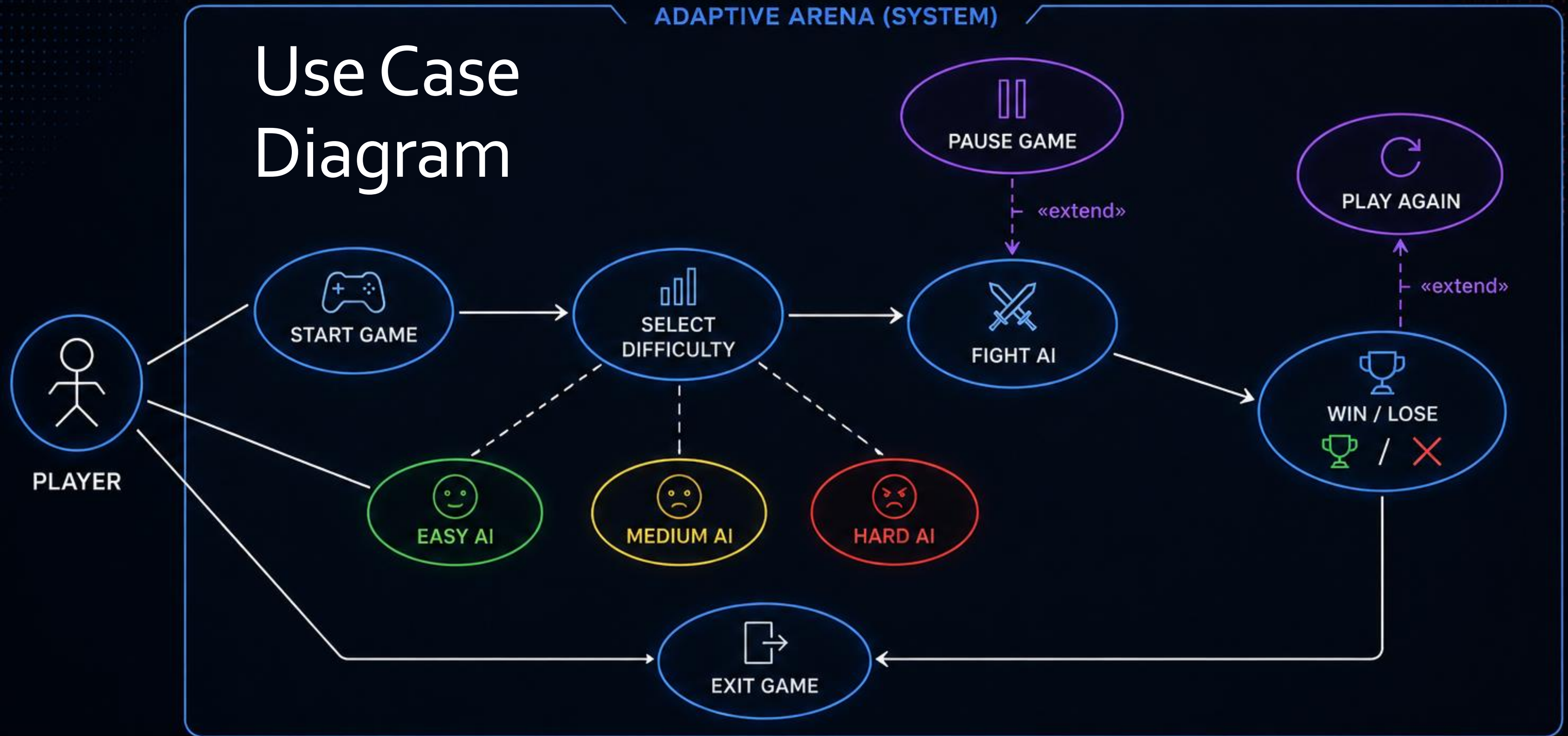
Class Diagram (UML Diagram)



ADAPTIVE ARENA

2D FIGHTING GAME

Use Case Diagram



Product Backlog

No.	User Stories	Acceptance Criteria	Feature
US_1	As a player, I want to access a main menu so that I can make choices in the game.	Main menu displays on game launch. Menu includes: Start, Settings, Quit, Credits. User can navigate menu using mouse. Selecting Start transitions to fight screen. Selecting Exit closes the game	Game Navigation
US_3	As a player, I want to be able to challenge an easy AI opponent so that I can practice basic gameplay mechanics.	An EasyController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack at random. The AI moves in the general direction of the player. The AI has a slower reaction update time, making its behavior less responsive and easier to predict.	Gameplay
US_4	As a player, I want to be able to challenge a medium AI opponent so that I can experience a balanced level of difficulty.	A MediumController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack with more consistency than random behavior. The AI moves toward the player more effectively than the easy AI. The AI has a moderate reaction update time, making it more responsive than easy but not overly difficult.	Gameplay
US_5	As a player, I want to be able to challenge a hard AI opponent so that I can face a highly challenging and intelligent opponent.	A HardController script is implemented using machine learning techniques. Unity ML-Agents are installed and configured with a working Python environment. The AI overrides CollectObservations, OnActionReceived, and OnEpisodeBegin methods. A reward and penalty system is implemented to guide AI learning. The AI is trained using a Python training process. The resulting trained neural network model is imported and used in the game. The hard AI demonstrates advanced and responsive behavior compared to easy and medium difficulties.	Gameplay
US_6	As a player, I want the player and AI characters to use pixel art sprites so that the characters match the retro visual style.	Player character uses a pixel art sprite. AI opponent uses a pixel art sprite. Sprites replace any placeholder shapes for characters and display correctly during gameplay.	Visual Design
US_7	As a player, I want to see a health bar during the fight so that I can track progress.	Player health bar is visible on screen during the match. AI health bar is also visible. Health bars decrease when damage is taken. Health bars update in real time during combat.	Combat UI
US_8	As a player, I want to win or lose based on health depletion so that the fight has a clear outcome.	When the AI's health reaches zero, the player wins. When the player's health reaches zero, the player loses. The match ends immediately when one health bar reaches zero. No further movement or attacks occur after the match ends.	Combat System

Product Backlog Continued

No.	User Stories	Acceptance Criteria	Feature
US_9	As a player, I want to see a win or lose screen so that I know the match outcome.	A win screen displays when the player wins. A lose screen displays when the player loses. End screen includes options to return to the main menu or play again. Selecting an option transitions properly without crashing the game.	Game Navigation
US_10	As a player, I want smooth transitions between game stages so that gameplay feels complete and polished.	Game transitions smoothly from main menu to difficulty selection and then to the match. Difficulty settings carry into the fight. After the match ends, the player can return to the main menu or restart the game without errors or crashes.	Game Architecture
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	Camera System
US_12	As a player, I want the game environment to use pixel art graphics so that the stage matches the retro aesthetic.	The game background uses pixel art design. Background assets load correctly during gameplay. The environment visuals are consistent with the pixel art style and no placeholder backgrounds appear in the Sprint demo.	Visual Design
US_13	As a player, I want character sprites to animate during movement and attacks so that the game feels dynamic and visually engaging.	Player sprites animate during movement and attacks. AI sprites animate during movement or attacks. Animations display smoothly during gameplay.	Visual Design
US_14	As a player, I want the ability to block attacks so that I can defend myself against enemy attacks during combat.	The player can activate a block action using a designated input. When blocking, incoming attacks deal reduced or no damage. A blocking animation or visual indicator appears while the player is blocking. Blocking only works while the block input is active.	Gameplay
US_15	As a player, I want the game to prevent my character from falling out of the environment so that gameplay stays within the intended boundaries.	The player character cannot move beyond the stage boundaries. The player cannot fall through the ground or platforms. Collision detection prevents the player from leaving the playable area. The player remains within the environment during gameplay.	Gameplay

Product Backlog Continued

No.	User Stories	Acceptance Criteria	Feature
US_16	As a player, I want the ability to perform a double jump so that I can reach higher areas and have greater mobility during gameplay.	The player can jump once from the ground and perform a second jump while in the air. The second jump can only occur after the first jump. The player cannot perform more than two jumps before landing. The double jump resets after the player lands on the ground.	Gameplay
US_17	As a player, I want to be able to use multiple control methods such as a gamepad controller or keyboard and mouse so that I can play the game using my preferred input device.	The game accepts input from a keyboard and mouse. The game accepts input from a gamepad controller. Core gameplay actions such as movement, jumping, attacking, and blocking can be performed using either control method. The game responds correctly to inputs from either device during gameplay.	Gameplay
US_18	As a player, I want sound effects and music in the game so that combat and menus feel more immersive and engaging.	Combat actions play sound effects during gameplay. Background music plays during menus or matches. UI interactions such as button selection and menu navigation play sound effects. Audio plays correctly without distortion or missing files.	Audio System
US_19	As a player, I want improved UI visuals, a settings page with volume controls, and credits so that the game feels polished and customizable.	UI art assets display correctly throughout menus and gameplay. Settings page allows the player to adjust volume levels. Volume settings affect game audio appropriately. Credits page displays team or project information correctly.	User Interface
US_20	As a player, I want additional combat actions such as duck, kick, and grab so that combat gameplay has more variety and strategy.	Player can perform a duck action using a designated input. Player can perform a kick attack during gameplay. Player can perform a grab action against opponents. New actions display proper animations and function during combat.	Combat System
US_21	As a player, I want the game to function without major bugs so that gameplay remains stable and enjoyable.	AI opponents perform intended combat actions during gameplay. Hard AI scoring and reward systems function correctly. Gameplay systems operate without crashes or game-breaking issues. Core mechanics work consistently during matches.	Quality Assurance

Product Backlog – Removed User Story

No.	User Story	Acceptance Criteria	Feature
US_2	As a player, I want to design my character so that I can play with a customized fighter.	Player can choose character name. Player can design character appearance. Character preview is displayed before confirmation. User confirms selection to proceed. Selected character loads into the match	Character System

Sprint 1 User Stories

No.	User Stories	Acceptance Criteria	Feature
US_1	As a player, I want to access a main menu so that I can make choices in the game.	Main menu displays on game launch. Menu includes: Start, Settings, Quit, Credits. User can navigate menu using mouse. Selecting Start transitions to fight screen. Selecting Exit closes the game	Game Navigation
US_7	As a player, I want to see a health bar during the fight so that I can track progress.	Player health bar is visible on screen during the match. AI health bar is also visible. Health bars decrease when damage is taken. Health bars update in real time during combat.	Combat UI
US_8	As a player, I want to win or lose based on health depletion so that the fight has a clear outcome.	When the AI's health reaches zero, the player wins. When the player's health reaches zero, the player loses. The match ends immediately when one health bar reaches zero. No further movement or attacks occur after the match ends.	Combat System
US_9	As a player, I want to see a win or lose screen so that I know the match outcome.	A win screen displays when the player wins. A lose screen displays when the player loses. End screen includes options to return to the main menu or play again. Selecting an option transitions properly without crashing the game.	Game Navigation

Sprint 1 User Stories Continued

No.	User Stories	Acceptance Criteria	Feature
US_14	As a player, I want the ability to block attacks so that I can defend myself against enemy attacks during combat.	The player can activate a block action using a designated input. When blocking, incoming attacks deal reduced or no damage. A blocking animation or visual indicator appears while the player is blocking. Blocking only works while the block input is active.	Gameplay
US_15	As a player, I want the game to prevent my character from falling out of the environment so that gameplay stays within the intended boundaries.	The player character cannot move beyond the stage boundaries. The player cannot fall through the ground or platforms. Collision detection prevents the player from leaving the playable area. The player remains within the environment during gameplay.	Gameplay
US_16	As a player, I want the ability to perform a double jump so that I can reach higher areas and have greater mobility during gameplay.	The player can jump once from the ground and perform a second jump while in the air. The second jump can only occur after the first jump. The player cannot perform more than two jumps before landing. The double jump resets after the player lands on the ground.	Gameplay

Sprint 1 Burndown Chart

QuestKeeper BurnChart

Current sprint

Previous sprints

⚙ Settings

Current Sprint Burndown

End sprint now



All

139 Completed

111 Added

111 Removed

+68 Reestimated

0 In progress

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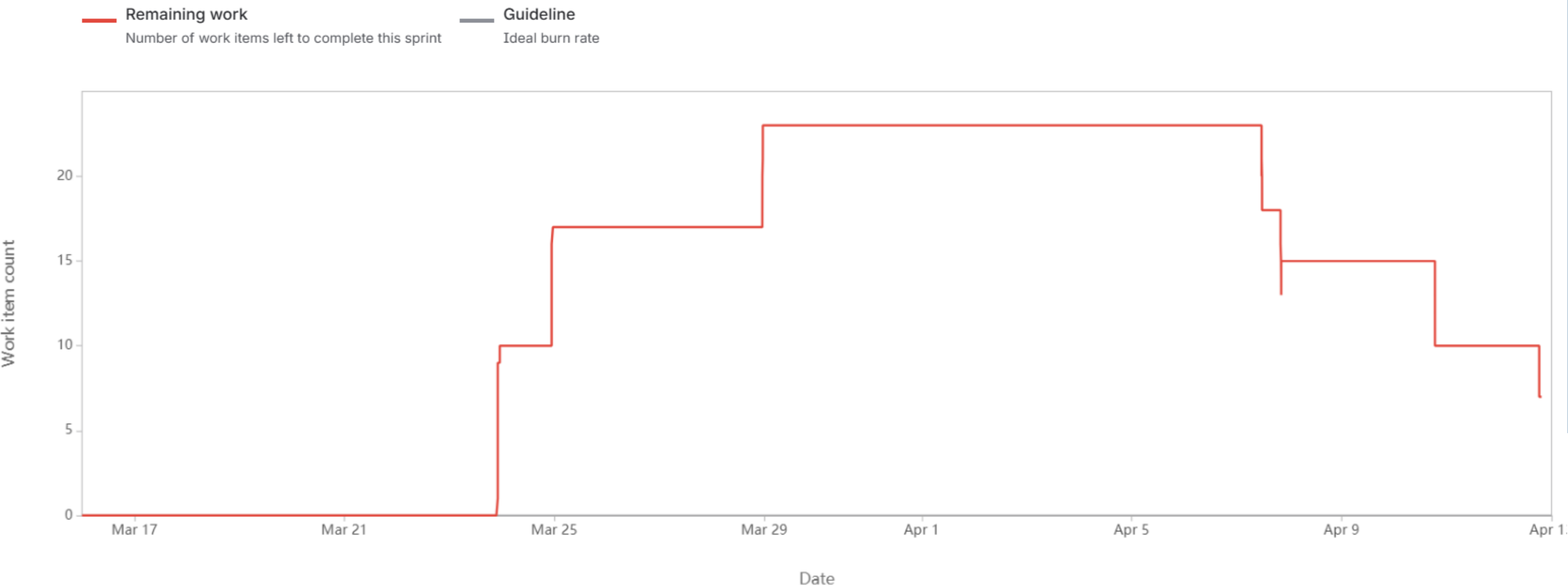
Sprint ▾

Sprint 2 User Stories

No.	User Stories	Acceptance Criteria	Feature
US_3	As a player, I want to be able to challenge an easy AI opponent so that I can practice basic gameplay mechanics.	An EasyController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack at random. The AI moves in the general direction of the player. The AI has a slower reaction update time, making its behavior less responsive and easier to predict.	Gameplay
US_4	As a player, I want to be able to challenge a medium AI opponent so that I can experience a balanced level of difficulty.	A MediumController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack with more consistency than random behavior. The AI moves toward the player more effectively than the easy AI. The AI has a moderate reaction update time, making it more responsive than easy but not overly difficult.	Gameplay
US_5	As a player, I want to be able to challenge a hard AI opponent so that I can face a highly challenging and intelligent opponent.	A HardController script is implemented using machine learning techniques. Unity ML-Agents are installed and configured with a working Python environment. The AI overrides CollectObservations, OnActionReceived, and OnEpisodeBegin methods. A reward and penalty system is implemented to guide AI learning. The AI is trained using a Python training process. The resulting trained neural network model is imported and used in the game. The hard AI demonstrates advanced and responsive behavior compared to easy and medium difficulties.	Gameplay

Sprint 2 Burndown Chart

Date - March 16th, 2026 - April 13th, 2026



Sprint 3 Stories Backlog

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint
US_6	As a player, I want the player and AI characters to use pixel art sprites so that the characters match the retro visual style.	Player character uses a pixel art sprite. AI opponent uses a pixel art sprite. Sprites replace any placeholder shapes for characters and display correctly during gameplay.	Visual Design	3	1	3
US_10	As a player, I want smooth transitions between game stages so that gameplay feels complete and polished.	Game transitions smoothly from main menu to difficulty selection and then to the match. Difficulty settings carry into the fight. After the match ends, the player can return to the main menu or restart the game without errors or crashes.	Game Architecture	4	3	3
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	Camera System	6 (2 already earned in sprint 1)	3	3
US_12	As a player, I want the game environment to use pixel art graphics so that the stage matches the retro aesthetic.	The game background uses pixel art design. Background assets load correctly during gameplay. The environment visuals are consistent with the pixel art style and no placeholder backgrounds appear in the Sprint demo.	Visual Design	3	1	3
US_13	As a player, I want character sprites to animate during movement and attacks so that the game feels dynamic and visually engaging.	Player sprites animate during movement and attacks. AI sprites animate during movement or attacks. Animations display smoothly during gameplay.	Visual Design	8	1	3

Sprint 3 Stories Backlog Continued

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint
US_17	As a player, I want to be able to use multiple control methods such as a gamepad controller or keyboard and mouse so that I can play the game using my preferred input device.	The game accepts input from a keyboard and mouse. The game accepts input from a gamepad controller. Core gameplay actions such as movement, jumping, attacking, and blocking can be performed using either control method. The game responds correctly to inputs from either device during gameplay.	Gameplay	5	2	3
US_18	As a player, I want sound effects and music in the game so that combat and menus feel more immersive and engaging.	Combat actions play sound effects during gameplay. Background music plays during menus or matches. UI interactions such as button selection and menu navigation play sound effects. Audio plays correctly without distortion or missing files.	Audio System	10	1	3
US_19	As a player, I want improved UI visuals, a settings page with volume controls, and credits so that the game feels polished and customizable.	UI art assets display correctly throughout menus and gameplay. Settings page allows the player to adjust volume levels. Volume settings affect game audio appropriately. Credits page displays team or project information correctly.	User Interface	7	1	3
US_20	As a player, I want additional combat actions such as duck, kick, and grab so that combat gameplay has more variety and strategy.	Player can perform a duck action using a designated input. Player can perform a kick attack during gameplay. Player can perform a grab action against opponents. New actions display proper animations and function during combat.	Combat System	9	1	3
US_21	As a player, I want the game to function without major bugs so that gameplay remains stable and enjoyable.	AI opponents perform intended combat actions during gameplay. Hard AI scoring and reward systems function correctly. Gameplay systems operate without crashes or game-breaking issues. Core mechanics work consistently during matches.	Quality Assurance	7	1	3

Sprint 3 Test Cases

No.	User Stories	Acceptance Criteria	Story Points		
US_06	As a player, I want the player and AI characters to use pixel art sprites so that the characters match the retro visual style.	Player character uses a pixel art sprite. AI opponent uses a pixel art sprite. Sprites replace any placeholder shapes for characters and display correctly during gameplay.	3		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_06_01	Player Character Uses Pixel Art Sprite	Launch the game and start a match. Observe the player character displayed on screen.	The player character appears as a pixel art sprite instead of a placeholder shape.	1	COMPLETE
TC_06_02	AI Character Uses Pixel Art Sprite	Launch the game and start a match against an AI opponent. Observe the AI character displayed on screen.	The AI character appears as a pixel art sprite instead of a placeholder shape.	1	COMPLETE
TC_06_03	Character Sprites Display Correctly During Gameplay	Start a match and play for several seconds while observing both characters.	Both player and AI characters consistently display pixel art sprites during gameplay without graphical issues.	1	COMPLETE
US_10	As a player, I want smooth transitions between game stages so that gameplay feels complete and polished.	Game transitions smoothly from main menu to difficulty selection and then to the match. Difficulty settings carry into the fight. After the match ends, the player can return to the main menu or restart the game without errors or crashes.	4		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_10_01	Difficulty Selection Menu Displays	Launch the game and navigate to the mode or match setup screen.	The player is presented with options to select easy, medium, or hard difficulty.	1	COMPLETE
TC_10_02	Selected Difficulty Persists Into Match	Launch the game, select any difficulty, and start a match. Observe gameplay settings.	The match runs using the selected difficulty without reverting or changing unexpectedly.	1	COMPLETE
TC_10_03	Difficulty Selection Loads Correct AI	Launch the game, select difficulty, and start a match. Observe AI behavior.	The AI behaves according to easy, medium, and hard difficulty.	2	COMPLETE
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	6 (2 already earned in sprint 1)		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_11_01	Camera Shows Both Fighters	Start a match and observe camera positioning during gameplay.	Both fighters remain visible on screen at the same time.	3	COMPLETE
TC_11_02	Camera Follows Fighter Movement	Move fighters across the stage to opposite ends.	Camera adjusts position to keep both fighters visible.	3	COMPLETE

Sprint 3 Test Cases Continued

No.	User Stories	Acceptance Criteria	Story Points		
US_12	As a player, I want the game environment to use pixel art graphics so that the stage matches the retro aesthetic.	The game background uses pixel art design. Background assets load correctly during gameplay. The environment visuals are consistent with the pixel art style and no placeholder backgrounds appear in the Sprint demo.	4		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_12_01	Pixel Art Background Loads in Match	Launch the game and start a match. Observe the stage background when gameplay begins.	The background environment loads successfully and displays pixel art graphics.	1	COMPLETE
TC_12_02	Environment Matches Pixel Art Style	Start a match and observe the environment and background elements during gameplay.	All environment visuals are designed in pixel art style consistent with the game's retro theme.	2	COMPLETE
TC_12_03	No Placeholder Background Appears	Start the demo and enter gameplay. Observe the background environment during the match.	No placeholder or unfinished background graphics appear.	1	COMPLETE
US_13	As a player, I want character sprites to animate during movement and attacks so that the game feels dynamic and visually engaging.	Player sprites animate during movement and attacks. AI sprites animate during movement or attacks. Animations display smoothly during gameplay.	8		
No.	Test Case Title	Test Flow	Expected Result	Execution Status	Execution Date
TC_13_01	Player Sprite Animates During Movement	Start a match and move the player character left or right across the stage.	The player sprite animates correctly while moving.	3	COMPLETE
TC_13_02	Player Sprite Animates During Attack	Start a match and perform an attack action with the player character.	The player sprite displays the correct attack animation when the action occurs.	3	COMPLETE
TC_13_03	AI Sprite Animates During Gameplay	Start a match and observe the AI opponent moving or attacking during gameplay.	The AI character sprite animates correctly during movement or attack actions.	2	COMPLETE
US_17	As a player, I want to be able to use multiple control methods such as a gamepad controller or keyboard and mouse so that I can play the game using my preferred input device.	The game accepts input from a keyboard and mouse. The game accepts input from a gamepad controller. Core gameplay actions such as movement, jumping, attacking, and blocking can be performed using either control method. The game responds correctly to inputs from either device during gameplay.	3		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_17_01	Keyboard and Mouse Controls Work	Launch the game and start a match. Use the keyboard and mouse to perform actions such as movement, jumping, attacking, and blocking.	The player character responds correctly to all keyboard and mouse inputs.	1	COMPLETE
TC_17_02	Gamepad Controller Controls Work	Launch the game with a connected gamepad controller and start a match. Use the controller to perform actions such as movement, jumping, attacking, and blocking.	The player character responds correctly to all controller inputs.	2	INCOMPLETE
TC_17_03	Game Responds to Active Input Device	Launch the game with both a controller and keyboard connected. Start a match and perform actions using each input device.	The game correctly registers inputs from either the keyboard and mouse or the controller during gameplay.	2	INCOMPLETE

Sprint 3 Test Cases Continued

No.	User Stories	Acceptance Criteria	Story Points		
US_18	As a player, I want sound effects and music in the game so that combat and menus feel more immersive and engaging.	Combat actions play sound effects during gameplay. Background music plays during menus or matches. UI interactions such as button selection and menu navigation play sound effects. Audio plays correctly without distortion or missing files.	10		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_18_01	Combat Sound Effects Play	Perform attacks during gameplay.	Combat actions produce sound effects correctly.	5	COMPLETE
TC_18_02	Background Music Plays	Launch the game and enter menus or matches.	Background music plays correctly during gameplay and menus.	2	COMPLETE
TC_18_03	UI Sound Effects Play	Navigate menus and select buttons.	UI interactions produce sound effects correctly.	3	COMPLETE
US_19	As a player, I want improved UI visuals, a settings page with volume controls, and credits so that the game feels polished and customizable.	UI art assets display correctly throughout menus and gameplay. Settings page allows the player to adjust volume levels. Volume settings affect game audio appropriately. Credits page displays team or project information correctly.	7		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_19_01	UI Art Displays Correctly	Launch the game and navigate menus.	UI art assets display properly without errors.	2	COMPLETE
TC_19_02	Volume Control Adjusts Audio	Open settings and modify volume controls.	Audio volume changes correctly according to settings.	3	COMPLETE
TC_19_03	Credits Display Correctly	Open the credits page from the main menu.	Credits information displays correctly.	2	COMPLETE
US_20	As a player, I want additional combat actions such as duck, kick, and grab so that combat gameplay has more variety and strategy.	Player can perform a duck action using a designated input. Player can perform a kick attack during gameplay. Player can perform a grab action against opponents. New actions display proper animations and function during combat.	9		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_20_01	Duck Action Functions	Use the duck input during gameplay.	Player performs duck action successfully.	2	COMPLETE
TC_20_02	Kick Action Functions	Use the kick input during gameplay.	Player performs kick attack successfully.	2	COMPLETE
TC_20_03	Grab Action Functions	Move near an opponent and activate grab input.	Player performs grab action successfully.	5	INCOMPLETE
US_21	As a player, I want the game to function without major bugs so that gameplay remains stable and enjoyable.	AI opponents perform intended combat actions during gameplay. Hard AI scoring and reward systems function correctly. Gameplay systems operate without crashes or game-breaking issues. Core mechanics work consistently during matches.	7		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_21_01	Medium AI Performs Combat Actions	Start a match against medium AI and observe gameplay.	Medium AI performs intended combat actions correctly.	2	COMPLETE
TC_21_02	Hard AI Uses Accurate Point System	Start a match against hard AI and observe scoring behavior.	Hard AI scoring and reward systems function correctly.	5	COMPLETE

Sprint 3 Stories Completed

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint
US_6	As a player, I want the player and AI characters to use pixel art sprites so that the characters match the retro visual style.	Player character uses a pixel art sprite. AI opponent uses a pixel art sprite. Sprites replace any placeholder shapes for characters and display correctly during gameplay.	Visual Design	3	1	3
US_10	As a player, I want smooth transitions between game stages so that gameplay feels complete and polished.	Game transitions smoothly from main menu to difficulty selection and then to the match. Difficulty settings carry into the fight. After the match ends, the player can return to the main menu or restart the game without errors or crashes.	Game Architecture	4	3	3
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	Camera System	6 (2 already earned in sprint 1)	3	3
US_12	As a player, I want the game environment to use pixel art graphics so that the stage matches the retro aesthetic.	The game background uses pixel art design. Background assets load correctly during gameplay. The environment visuals are consistent with the pixel art style and no placeholder backgrounds appear in the Sprint demo.	Visual Design	3	1	3
US_13	As a player, I want character sprites to animate during movement and attacks so that the game feels dynamic and visually engaging.	Player sprites animate during movement and attacks. AI sprites animate during movement or attacks. Animations display smoothly during gameplay.	Visual Design	8	1	3
US_18	As a player, I want sound effects and music in the game so that combat and menus feel more immersive and engaging.	Combat actions play sound effects during gameplay. Background music plays during menus or matches. UI interactions such as button selection and menu navigation play sound effects. Audio plays correctly without distortion or missing files.	Audio System	10	1	3
US_19	As a player, I want improved UI visuals, a settings page with volume controls, and credits so that the game feels polished and customizable.	UI art assets display correctly throughout menus and gameplay. Settings page allows the player to adjust volume levels. Volume settings affect game audio appropriately. Credits page displays team or project information correctly.	User Interface	7	1	3
US_21	As a player, I want the game to function without major bugs so that gameplay remains stable and enjoyable.	AI opponents perform intended combat actions during gameplay. Hard AI scoring and reward systems function correctly. Gameplay systems operate without crashes or game-breaking issues. Core mechanics work consistently during matches.	Quality Assurance	7	1	3

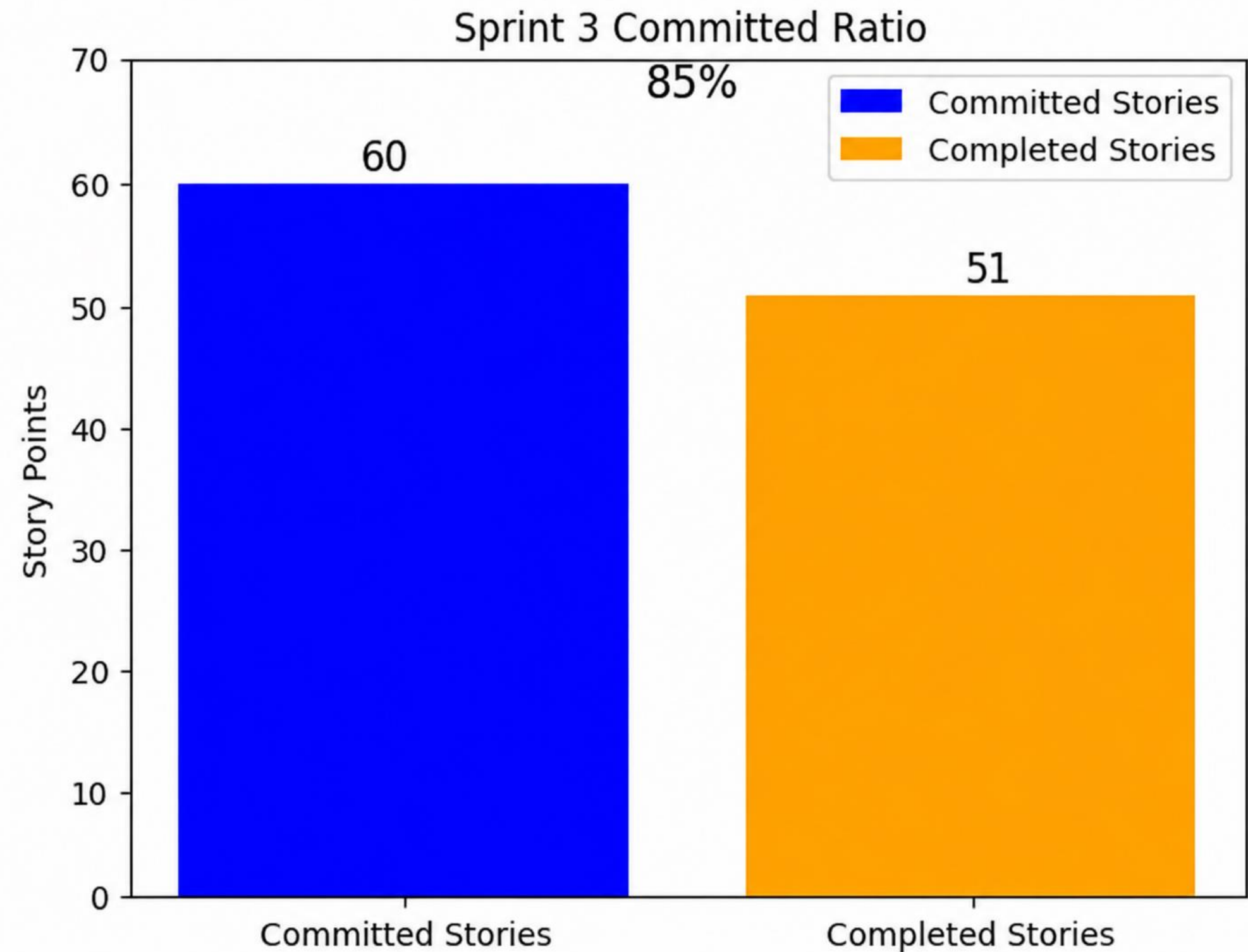
Sprint 3 Stories Not Completed

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint
US_17	As a player, I want to be able to use multiple control methods such as a gamepad controller or keyboard and mouse so that I can play the game using my preferred input device.	The game accepts input from a keyboard and mouse. The game accepts input from a gamepad controller. Core gameplay actions such as movement, jumping, attacking, and blocking can be performed using either control method. The game responds correctly to inputs from either device during gameplay.	Gameplay	5	2	3
US_20	As a player, I want additional combat actions such as duck, kick, and grab so that combat gameplay has more variety and strategy.	Player can perform a duck action using a designated input. Player can perform a kick attack during gameplay. Player can perform a grab action against opponents. New actions display proper animations and function during combat.	Combat System	9	1	3

Team Velocity – Sprint 3

Story Points Committed: 60

Story Points Completed: 51



Team Velocity

Historical ratios:

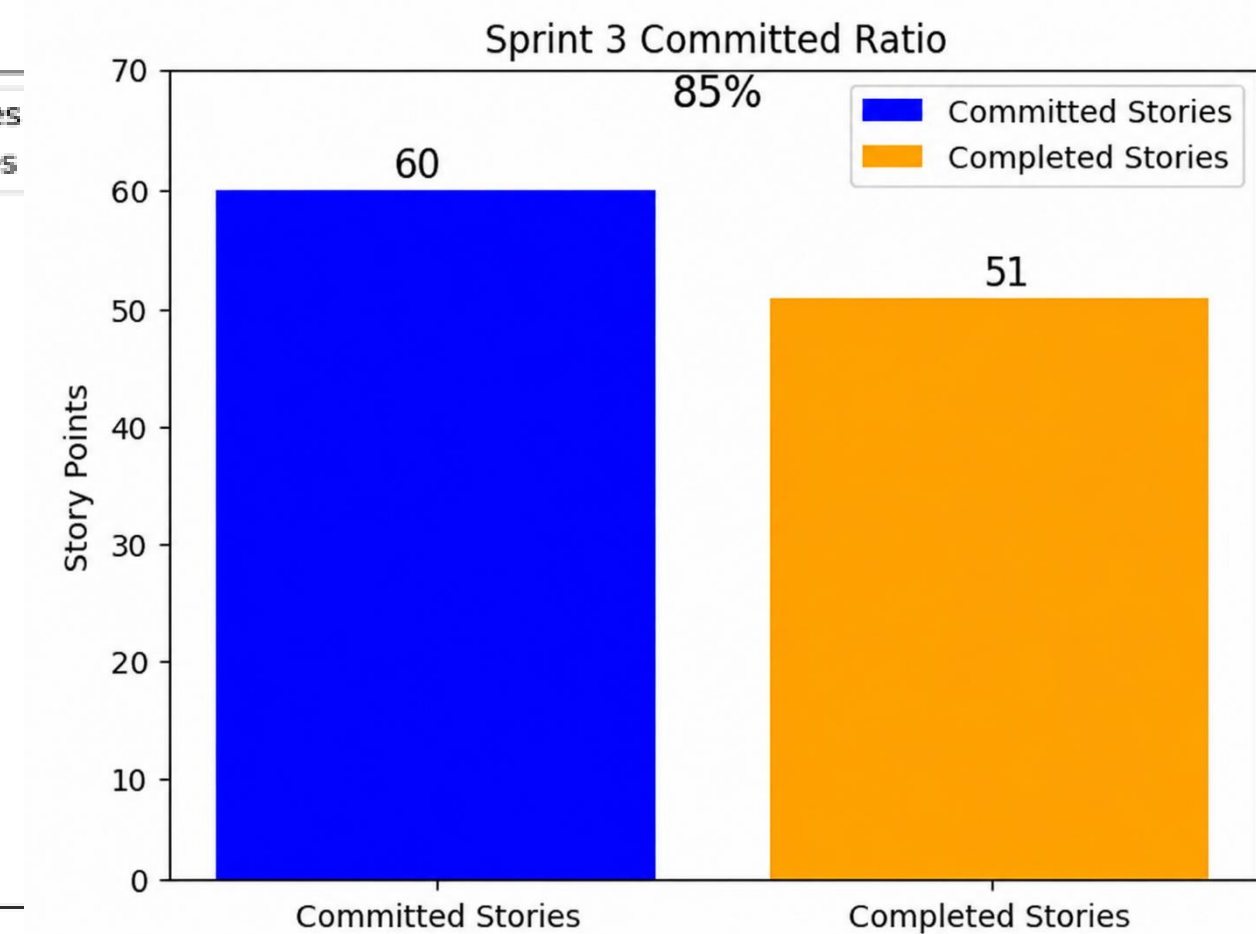
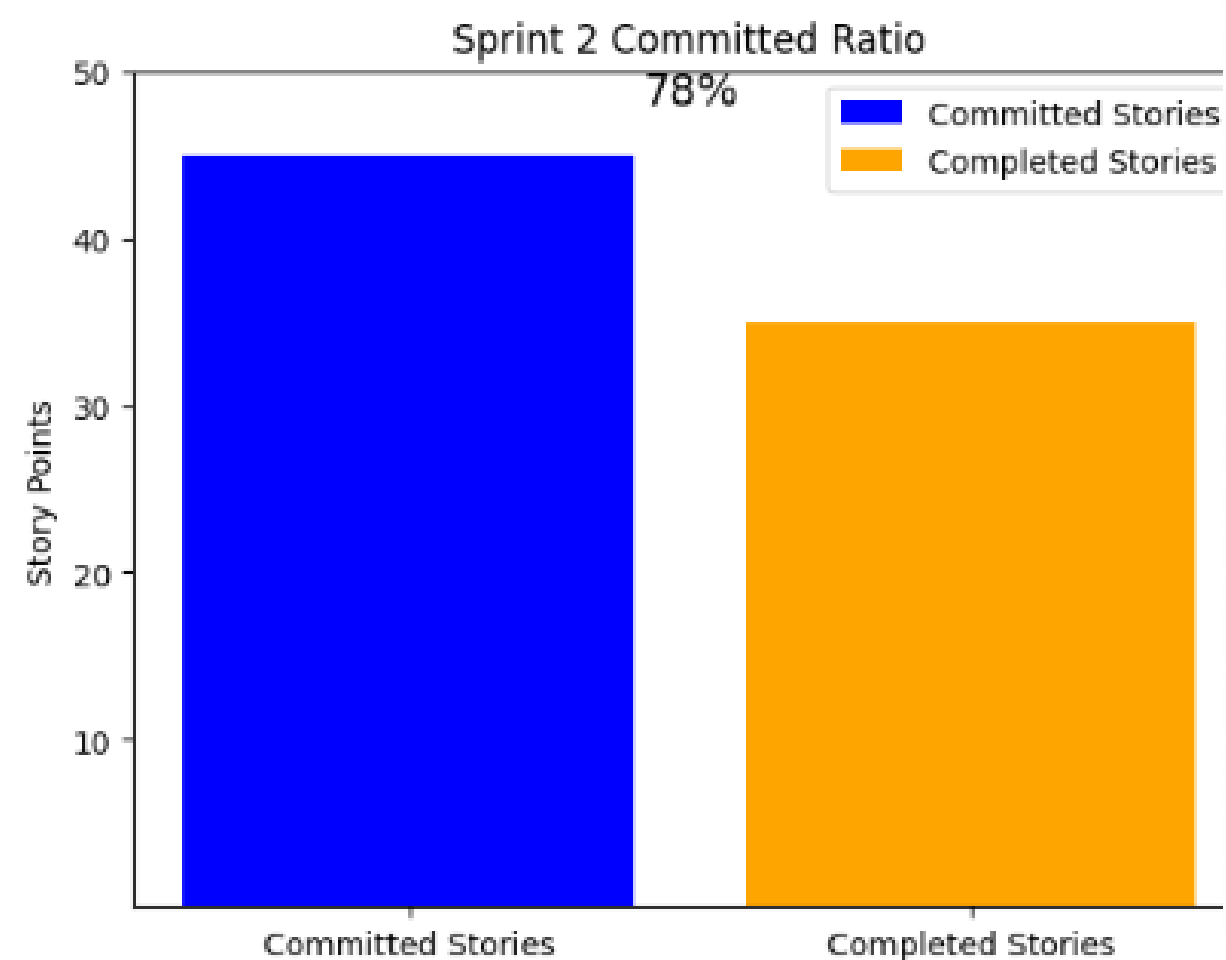
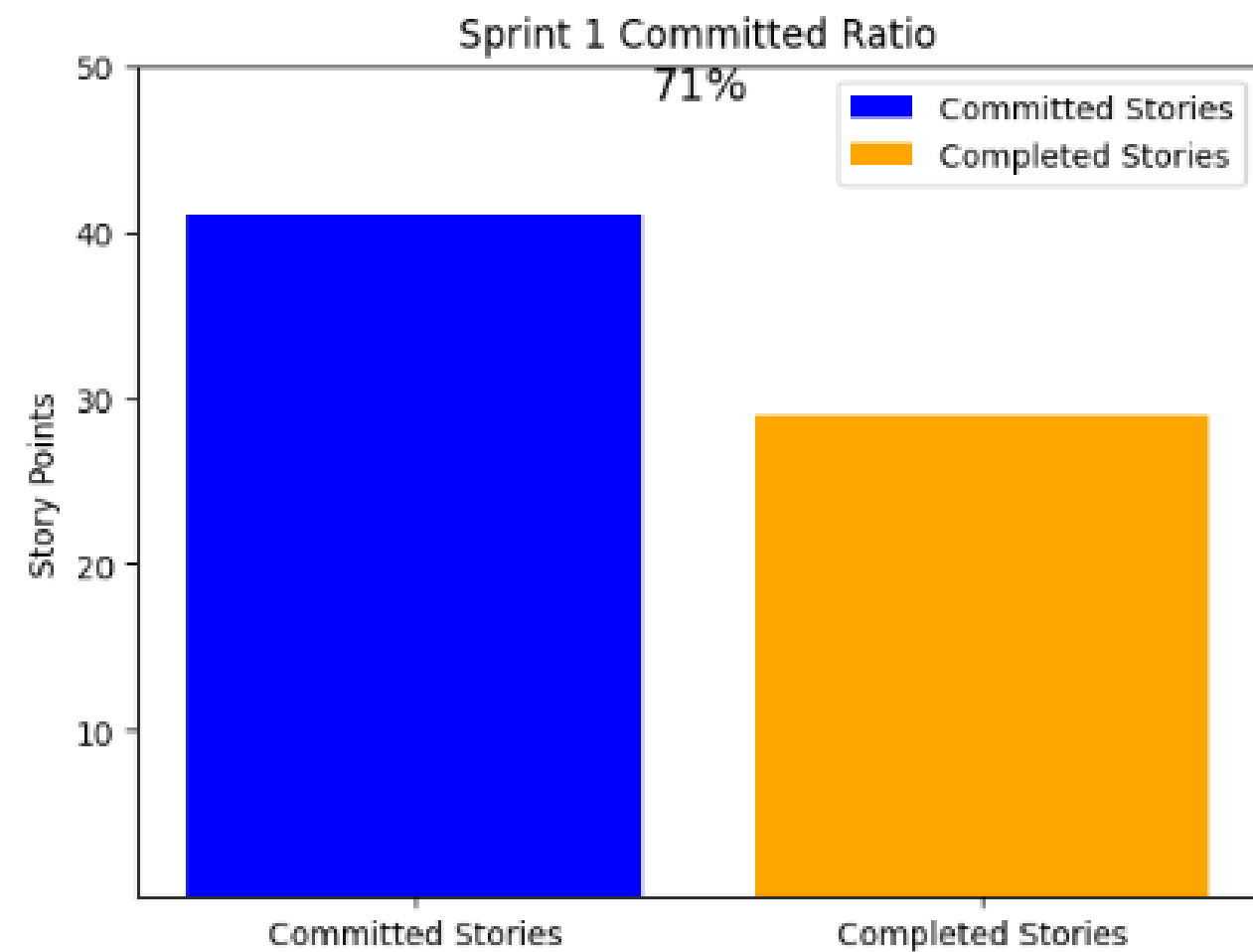
- Sprint 1: 70.73%
- Sprint 2: 77.78%
- Sprint 3: 85%

Improvement:

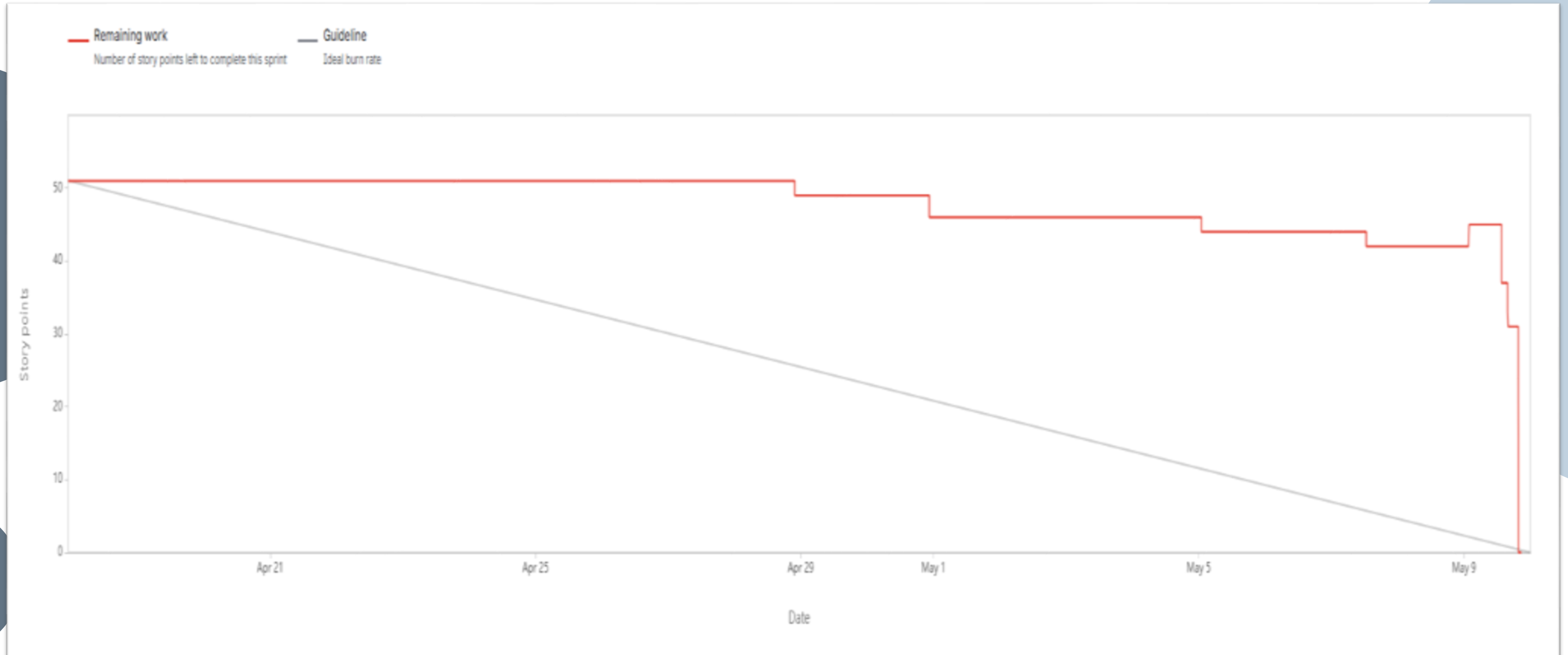
- +7.05%, 7.22%

Average ratio:

- 77.84%



Sprint 3 Burndown Chart



Retrospective

QuestKeeper Retrospective Sprint 3

What went well +

The team collaborated effectively throughout the sprint

+0

Maintained consistent development velocity, resulting in high-quality deliverables that meet project requirements

+0

Improvement in technical execution and code stability compared to previous sprints

+0

Utilized Slack channels effectively for streamlined team communication and rapid problem-solving.

+0

Utilized Slack channels effectively for streamlined team communication and rapid problem-solving.

Each team member understood their individual responsibilities thanks to well-defined User Stories and tasks

+0

We successfully implemented a user-friendly interface for our players

+0

Starting the Sprint 3 planning meeting as early as possible helped us define a solid list of must-complete tasks.

+0

We need to ensure that our work status is updated consistently throughout the sprint to avoid information gaps

+0

We need to dedicate more time to studying the integration between Unity and GitHub to prevent technical conflicts

+0

We should utilize Jira more effectively to maintain clear visibility of our current progress and task status

The team needs to become more familiar with GitHub workflows to manage version control and code integration more smoothly

We must aim to finish core tasks earlier to provide extra time for quality assurance and final adjustments

+0

What can be improved +

Action Items +

Increase collaboration while coding to solve Unity-related Git issues in real-time

+0

Create a detailed list of tasks and subtasks with specific deadlines for the upcoming sprint

+0

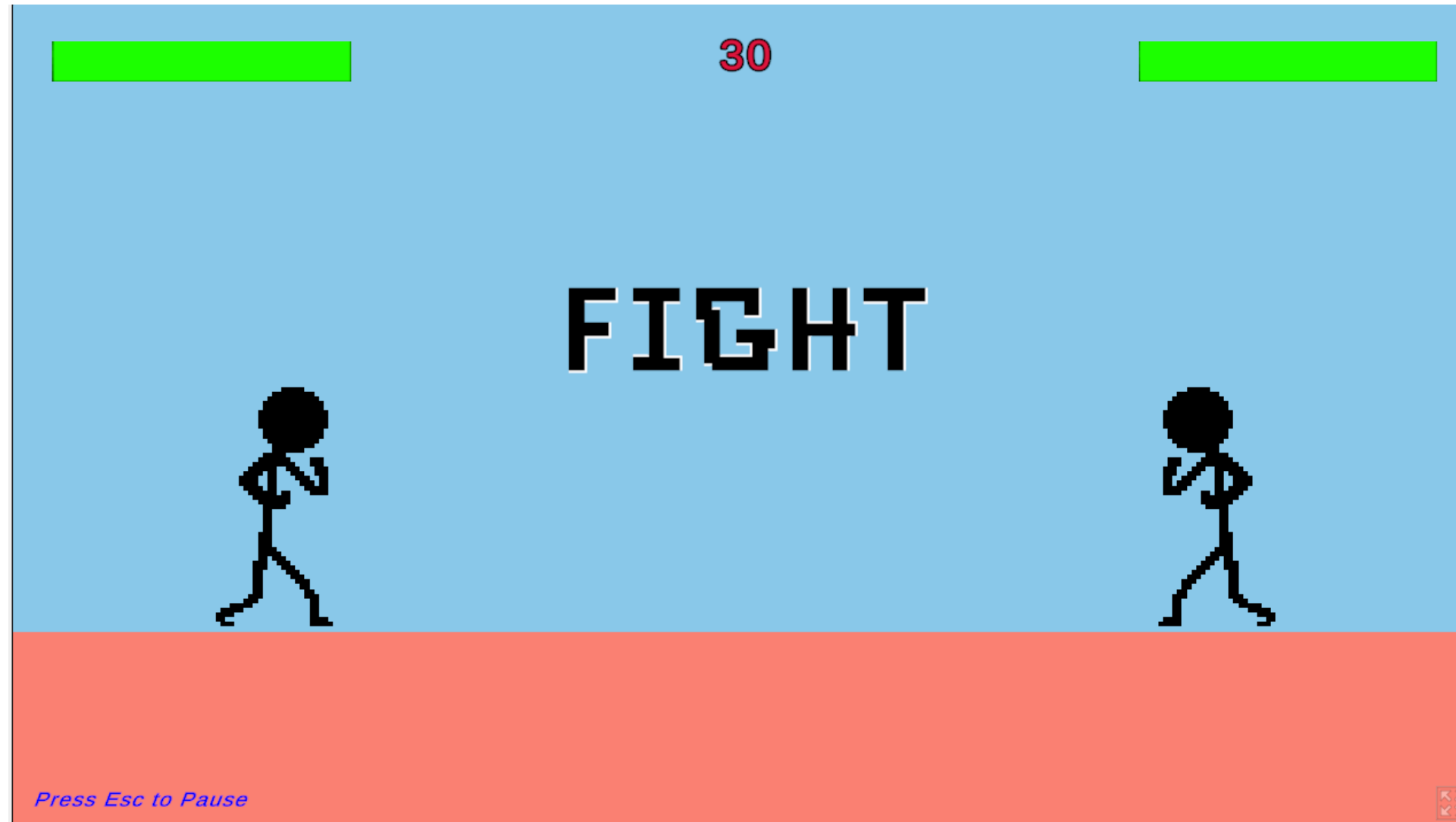
Conduct playtests with friends to gather feedback before the final project submission

+0

Schedule dedicated time for the team to get familiar with GitHub and resolve version control issues

+0

Project Demo



Adaptive Arena Demo



Wiki Page Link

<https://github.com/htmw/2026SA-QuestKeeper/wiki>





**Thank
You**